1.3 Linear Algebra: Problems

Solutions are in solutions.md. Attempt each problem seriously for at least 20–30 minutes before you look. $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$ unless stated otherwise, and all spaces are finite-dimensional unless stated otherwise.

Warm-up

1. Decide which of the following are subspaces, and justify each answer.

(a) $\{x \in \mathbb{R}^3 : x_1 + 2x_2 - x_3 = 0\}$.
(b) $\{x \in \mathbb{R}^3 : x_1 x_2 = 0\}$.
(c) $\{x \in \mathbb{R}^3 : x_1 = 1\}$.
(d) $\{f \in C[0,1] : f(0) = 2f(1)\}$, as a subset of $C[0,1]$.

2. Show that $(1,1,0), (1,0,1), (0,1,1)$ is a basis of $\mathbb{R}^3$. Is it a basis of $\mathbb{F}_2^3$, where $\mathbb{F}_2$ is the field with two elements?

3. Let $D : \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be differentiation. Write the matrix of D with respect to the basis $1, x, x^2, x^3$. Find $\operatorname{null} D$, $\operatorname{range} D$, the minimal polynomial of D and the Jordan form of D.

4. Let A_n be the $n \times n$ tridiagonal matrix with 2 on the diagonal and 1 on the sub- and super-diagonals. Compute $\det A_4$, then find $\det A_n$ for general n.

5. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Find the eigenvalues and eigenvectors, diagonalise A, and give a closed formula for A^n.

6. On $\mathcal{P}_2(\mathbb{R})$ with $\langle p, q \rangle = \int_{-1}^{1} p(x)q(x)\,dx$, apply Gram–Schmidt to $1, x, x^2$.

Core

7. (a) Prove that if $\dim V > \dim W$, no linear map $V \to W$ is injective.

(b) Let U, W be subspaces of V. By applying rank–nullity to the map $U \times W \to V$, $(u, w) \mapsto u + w$, prove that $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

8. Let $P \in \mathcal{L}(V)$ satisfy $P^2 = P$. Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$ and that P is diagonalisable. What are its possible eigenvalues?

9. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x+y,\ 2y)$. Find the matrix of T with respect to the standard basis, then with respect to $\mathcal{B}' = ((1,0),(1,1))$. Do this both directly and with the formula $P^{-1}AP$.

10. Let $T \in \mathcal{L}(V, W)$. Prove that $\operatorname{null} T' = (\operatorname{range} T)^0$, and deduce that $\dim \operatorname{range} T' = \dim \operatorname{range} T$. Explain why this shows that row rank equals column rank.

11. Find the characteristic and minimal polynomials of

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Which of them is diagonalisable? Are A and B similar?

12. Let $T \in \mathcal{L}(V)$ over $\mathbb{R}$ satisfy $T^3 = T$. Prove that T is diagonalisable and that its eigenvalues lie in $\{-1, 0, 1\}$.

13. Prove that eigenvectors of T corresponding to distinct eigenvalues are linearly independent.

14. Let T be a self-adjoint operator on an inner product space. Prove that every eigenvalue of T is real, and that eigenvectors for distinct eigenvalues are orthogonal.

15. Find an orthogonal matrix Q and a diagonal D with $A = QDQ^t$, where

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

16. Find the Jordan form J of

$$A = \begin{pmatrix} 5 & 4 & 2 & 1 \\ 0 & 1 & -1 & -1 \\ -1 & -1 & 3 & 0 \\ 1 & 1 & -1 & 2 \end{pmatrix},$$

and an invertible P with $P^{-1}AP = J$. (Hint: $\chi_A(z) = (z-1)(z-2)(z-4)^2$.)

17. Let $V = \mathcal{P}_2(\mathbb{R})$ and let $\mathrm{ev}_a \in V'$ be $p \mapsto p(a)$.

(a) Show that $\mathrm{ev}_0, \mathrm{ev}_1, \mathrm{ev}_2$ is a basis of V'.
(b) Find the basis p_0, p_1, p_2 of V whose dual basis is $\mathrm{ev}_0, \mathrm{ev}_1, \mathrm{ev}_2$.
(c) Use (b) to write down the unique $p \in V$ with $p(0) = 1$, $p(1) = 3$, $p(2) = 7$.

18. Let $U \subseteq V$ be a subspace and $T \in \mathcal{L}(V, W)$.

(a) Prove that $\dim V/U = \dim V - \dim U$.
(b) Prove that $V/\operatorname{null} T \cong \operatorname{range} T$. Check carefully that your map is well defined.
(c) Prove that if $U \subseteq \operatorname{null} T$, then T induces a linear map $V/U \to W$.

19. Let $A \in \mathbb{F}^{2\times 2}$ have trace t and determinant d. Verify directly that $A^2 - tA + dI = 0$. Use this to compute A^{-1} for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ without the adjugate formula.

Challenge

20. Prove, without determinants, that every operator on a nonzero finite-dimensional complex vector space has an eigenvalue. Where does the proof use $\mathbb{C}$?

21. Let $S, T \in \mathcal{L}(V)$ be diagonalisable with $ST = TS$. Prove that there is a basis of V consisting of vectors that are eigenvectors of both S and T.

22. (Schur.) Let V be a complex inner product space and $T \in \mathcal{L}(V)$. Prove that T has an upper-triangular matrix with respect to some orthonormal basis. You may assume T is upper triangular with respect to *some* basis.

23. (a) For a nilpotent N on $\mathbb{C}^n$, explain why the minimal polynomial gives the largest Jordan block and $\dim \operatorname{null} N$ gives the number of blocks.

(b) Find two 7×7 nilpotent matrices that have the same minimal polynomial and the same nullity but are **not** similar. Prove they are not similar.

24. (a) Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for $A, B \in \mathbb{F}^{n\times n}$, and deduce that there are no $A, B \in \mathbb{R}^{n\times n}$ with $AB - BA = I$.

(b) Let p be prime and $V = \mathbb{F}_p[x]/(x^p)$, with basis $1, x, \ldots, x^{p-1}$. Show that differentiation D and multiplication by x (X) are well-defined operators on V with $DX - XD = I$. Why does this not contradict (a)?